1.4.4 Statistics and Medicine

In medical field, statistical methods are extensively used. If we look at the medical journals one can understand to what extent the statistical techniques play a key role. Medical statistics deals with the applications of statistical methods like tests of significance and confidence intervals to medicine and health science including epidemiology, public health. Modern statistical methods helps the medical practitioners to understand how long a patient affected by a dreaded disease will survive and what are the factors that influence a patient to be alive or dead.



A Digital Revolution in Healthcare

- The Economist, Feb 03,2018

We know that the internet already enables patients to seek online consultations when and where it suits them. Now technologies such as the smartphones allow people to monitor their own health.

The possibilities multiply when you add to crucial missing ingredients- access to your own medical records and the ability, easily to share information with those you trust. That allows you to reduce inefficiencies in your own treatment and also to provide date to help train medical algorithms. You can enhance your own care and everyone else's too.

1.4.5 Statistics and Agriculture

Experimentation and inference based on these experiments are the key features of general scientific methodology. Agricultural scientists conduct experiments and make inferences to decide whether the particular variety of crop gives a better yield than others or a particular type of fertilizer etc,. There are several institutes where research is being done by making use of statistical methods like analysis of variance (ANOVA), factorial experiments etc., falls under the hut of Design of experiments. There is a separate institute (IASRI), New Delhi, carrying out research in agricultural statistics.

1.4.6 Statistics and Industry

Statistical methods play a vital role in any modern use of science and technology. Many statistical methods have been developed and applied in industries for various problems. For example, to maintain the quality of manufactured products the concept of statistical quality control is used. The quality in time domain study of mechanical, electrical or electronic items the concept of 'Reliability' has emerged. Total quality management and six-sigma theories make use of statistical concepts.

1.4.7 Statistics and Information Technology

Information Technology is the applications of computers and telecommunication equipments to store, retrieve, transmit and manipulate data. Now-a-days, several industries are involved in information technology and massive amounts of data are stored every day. These data are to be analyzed meaningfully so that the information contained in the data

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